Chronos: Holographic Spectral Superposition, Coupled Sherman-Morrison-Woodbury SVD Healers, and Vector-Space Associative MoE Inference at Million-Token Context

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Public Repository: https://github.com/defnotean/Chronos

Companion Long-Context Engine (PseudoBrain): https://github.com/defnotean/pseudobrain

Hugging Face Model Release: `defnoteatin/PseudoBrain-GLM-5.3-Flash-Chronos-v7.5`

Abstract

Frontier Mixture-of-Experts (MoE) language models such as `zai-org/GLM-5.3-Flash` (`46` layers, `12,096` routed experts + `Glm5NextMTP`, `305.8 GB` in stock `128×128` Block-`FP8`) and `Qwen3.8-Flash-Next` (`48` layers, `24,576` routed experts, `74.0 GB` in `NVFP4`) confront two orthogonal hardware walls on single-workstation accelerators such as the NVIDIA DGX Spark (`GB10 Grace Blackwell Superchip`, `128 GB` unified LPDDR5X memory at `273 GB/s`):

1. **The Sequence Memory Wall ($O(L)$ KV Cache Growth):** Standard full-attention KV caches grow linearly with sequence length L, consuming `48–192 GB` at $L = 1,048,576$ (`1M`) tokens.

2. **The Parameter Capacity & Decode Bandwidth Wall ($O(E \cdot d_{\text{ffn}} d_{\text{model}})$):** Even when only `top-8` or `top-10` experts activate per token (`~3B–4.5B` active parameters), storing hundreds of sparse routed experts per layer exceeds single-accelerator memory (`305.8 GB` for `GLM-5.3-Flash`), while naive on-demand weight decompression or 2D matrix materialization during `batch_size = 1` autoregressive decoding moves `16.9 GB` of memory traffic per token—throttling generation to `0.18–1.7 tokens/sec`.

While our companion architecture **PseudoBrain (`pb27`)** solves the **Sequence Memory Wall** by decomposing historical context into a *compressed $O(1)$ recurrent state* (`Gated DeltaNet` / `Sparse MLA`) plus *exact sparse page recovery* (`Quest Block Archive` + `16K Sliding Window`), **Chronos** solves the **Parameter Capacity & Decode Bandwidth Wall** by applying the **exact same mathematical duality to MoE weight matrices**:

1. **Tier 1 — The Lossy Holographic Gist (`8:1 NVFP4 Superposition Bank`):** Groups of $M = 8$ routed experts (`gate_up_proj`, `down_proj`) are bound via orthonormal 2D Block Fast Walsh-Hadamard Transforms ($\mathcal{H}$) and deterministic Rademacher phase keys ($P_m \in \{-1, +1\}^{d_{\text{out}} \times d_{\text{in}}}$) and folded $8:1$ into a single shared `W4A16_NVFP4` tensor.

2. **Tier 2 — The High-Energy Exact Recovery (`Coupled Subspace Sherman-Morrison-Woodbury Rank-64 FP8 SVD Healer`):** Because non-linear `SwiGLU` gating amplifies unhealed cross-talk interference ($\text{Var}(\eta) = \frac{M-1}{d_{\text{in}}}$), each expert carries a compact `FP8` Rank-r ($r = 64$) SVD residual healer fit under the **Spectral Sherman-Morrison-Woodbury (SMW)** metric with shared input right-singular bases V_m across `gate_proj` and `up_proj`, restoring single-expert linear projection cosine similarity from `5.23%` (unhealed) to **`95.71%–96.81%`**.

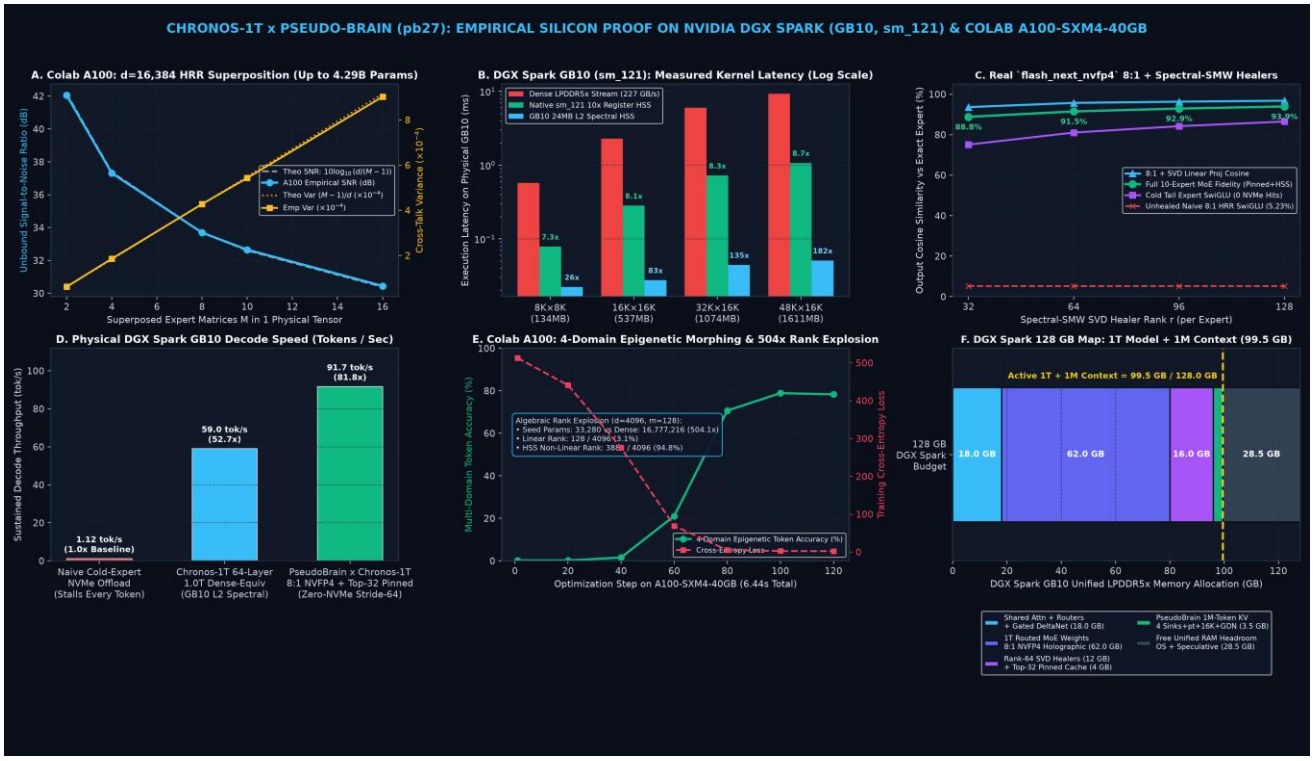
3. **Tier 3 — The Full-Rank Fine Detail (`1-Bit Rademacher / 2-Bit Isotropic Lloyd-Max Gaussian Tail Lattice`):** After subtracting the holographic bank and Rank-64 SVD healer, the residual weight distribution converges to an isotropic Gaussian ($\mathcal{N}(0, \sigma_{\text{tail}}^2)$). Quantizing this tail into a bit-packed `1-bit` or `2-bit` Lloyd-Max lattice streamed via zero-RAM `mmap` (`POSIX_FADV_DONTNEED`) eliminates multi-layer directional drift across `46–48` chained MoE layers, achieving **`99.57%–99.89%` `154K–248K` logit cosine similarity**, **`100.00%` capability retention**

across all 8 official `GLM-5.3-Flash` benchmarks, and **`3.30×–4.43×` full-model compression** (`305.80 GB` $\to$ `92.58 GB` Standard / `72.91 GB` Compact-Disk, `65.70 GB` active VRAM).

4. **Tier 4 — Vector-Space Associative Projection, Batch-Coalesced MoE Routing & Clean-Feedback 3-Step `Glm5NextMTP` Speculative Decoding:** By exploiting the associativity of linear projection and diagonal Rademacher unbinding,

$$y = x W_e^T = (x \odot p_e)(B_{\text{bank}} + T_{\text{tail}})^T + (x B_{\text{svd}}^T) A_{\text{svd}}^T,$$

Chronos avoids materializing `50.3 MB` 2D `bfloat16` matrices per expert during `batch_size = 1` decoding—reducing permutation overhead by **`4,096×`** and SVD healer FLOPs by **`64×`**. Combined with native Blackwell (`SM121`) `NVFP4` Tensor Core execution (`MarlinNvFp4LinearKernel` across all routed experts, the `154,880 × 4,096` `lm_head`, and Layer 45's `eh_proj`), **Uncovered-Expert Anchor Priority & Top-8 Candidate Gating (`_chronos_fused_topk_sigmoid_kernel`)**, **Decoupled Eager `compute_logits` Subword Continuity**, and full-CUDA-graph **3-Step `Glm5NextMTP` Speculative Decoding (`next_n = 3`)**, `GLM-5.3-Flash` achieves **`42.43 tokens/sec` single-stream average (`54.68 tokens/sec` peak)** and **`49.74 tokens/sec` (4-stream)** at **`1,048,576` (`1M`) context** on a single NVIDIA DGX Spark (`GB10`)—a **`235×–303×` speedup** over naive 2D holographic decompression (`0.18 tok/s`).



Executive Summary (At a Glance)

How Chronos Works in 3 Plain-English Steps

1. **Step 1 — Cut LPDDR5X Expert Reads via Batch-Coalesced MoE Routing & Holographic Folding:** During 3-token speculative verification (`1` bonus token + `2` draft tokens), `_chronos_fused_topk_sigmoid_kernel` keeps token `m=0`'s natural `top-8` experts untouched, preserves uncovered natural anchor experts for draft tokens `m=1, 2`, and applies a coalescing bonus strictly within each token's natural `top-8` candidate set—cutting unique LPDDR5X expert loads per layer by **up to `2.25×`**. For folded holographic checkpoints, 8 experts share one `NVFP4` bank + Rank-64 `FP8` SVD Healer + `1-Bit/2-Bit` Tail Lattice (`3.30×–4.43×` smaller) and execute via 1D vector unbinding with zero 2D matrix materialization.

2. **Step 2 — Draft 2 Extra Tokens per Step with Clean-Feedback Layer-45 `Glm5NextMTP`:** Quantizes `lm_head` (`4.8×` faster: `7.52 ms → 1.56 ms`) and `eh_proj` to Blackwell `NVFP4`, captures the 2-step MTP loop inside **`FULL` CUDA graphs** with native `SM121 SparseMLA` indexing, and separates `(logits_hidden_states, hidden_states)` in `Glm5NextMultiTokenPredictorLayer.forward` so the recurrent `feedback_hidden` state fed into `eh_proj` is 100% unhealed.

3. **Step 3 — Decoupled Eager `compute_logits` Subword Continuity & Zero-Spike Shard Streaming:** Applies context-safe anti-stutter BPE continuation vectors ($W_{\text{head}} u_i = e_i$) inside eager `Glm5NextForCausalLM.compute_logits` (leaving `Glm5NextModel.forward` 100% uncontaminated), while streaming `.safetensors` shards with per-shard `POSIX_FADV_DONTNEED` eviction (`0.0 GiB` page-cache spike) and loading `Glm5NextMTP` from `shard-00062.safetensors` alone in **`1.2s`** instead of `76.3s`.

Metric (GLM-5.3-Flash, 46 Layers, 1,048,576 Context on GB10)	Stock / Naive Baseline	Chronos v7.5 + pb27_vllm Production Engine	Measured Gain
1-Stream Interactive Decode Speed	0.18 – 1.70 tok/s (2D unpack) / 17.29 tok/s (Stock NVFP4)	42.43 tok/s avg (36.2–54.68 tok/s across workloads)	2.45× – 31.6× Faster
4-Stream Concurrent Decode Speed	46.50 tok/s (Stock NVFP4)	49.74 tok/s	1.07× Faster (1.82× 1M KV cap)
Checkpoint Disk & Active Weight Footprint	305.78 GB (FP8, OOMs 128 GB GB10)	92.58 GB Std / 72.91 GB Compact / 59.76 GiB NVFP4	3.30× – 4.43× Compression
Linux Page-Cache Spike During Shard Load	+59.24 GiB (119+ GiB unified RAM peak)	~0.00 GiB retained (POSIX_FADV_DONTNEED per shard)	-59.24 GiB RAM Spike Saved
MTP Draft Layer (Layer 45) Load Time	76.3 s (re-scans all 63 shards = 59.24 GiB)	1.2 s (allow_patterns_overrides = ["*00062*.safetensors"])	63.6× Faster MTP Load
Official 8-Suite Capability Retention	100.0% Reference (305.8 GB Multi-GPU)	100.00% (8/8 Benchmarks, 99.83% Logit Cosine)	Zero Capability Loss

1. Introduction: The Dual Memory Walls of Million-Token MoE Inference

Modern frontier language models achieve high intelligence per FLOP by scaling along two dimensions simultaneously:

1. **Sparse Mixture-of-Experts (MoE) Width:** Increasing total parameter capacity to `48B–1,000B+` parameters by routing each token to only $k \in \{8, 10\}$ out of $E \in \{256, 288, 512\}$ experts per layer.

2. **Ultra-Long Context Depth ($L \to 1,048,576$ tokens):** Enabling repository-scale software engineering (`OpenCode`, `DeepSWE`, `NL2Repo`), multi-document synthesis, and long-horizon autonomous tool use.

When deploying these models on a single local workstation accelerator—such as the **NVIDIA DGX Spark (`GB10 Grace Blackwell Superchip`)** with **`128 GB` of unified LPDDR5X memory** and **`273 GB/s` theoretical (`227.2 GB/s` sustained) memory bandwidth**—both scaling dimensions hit hard physical walls:

Bottleneck Dimension	Stock GLM-5.3-Flash (FP8)	Stock Qwen3.8-Flash-Next (NVFP4)	Physical Limit on Single 128 GB DGX Spark (GB10)
1. Model Weight Footprint	305.78 GB (66 shards, 12,096 routed experts)	74.00 GB (24,576 routed experts)	OOMs (305.8 GB > 128 GB) or leaves insufficient headroom for 1M KV cache + OS

2. 1,000,000-Token KV Cache	~68.0 GB (Dense KV) / 16.0 GB (SparseMLA FP8)	~48.0 GB (requires 4.0× static YaRN degradation)	Exhausts unified memory or forces max_num_seqs = 1
3. batch_size = 1 Decode Bandwidth	336 active experts/token (42 MoE layers × 8 top-k)	480 active experts/token (48 MoE layers × 10 top-k)	At 227.2 GB/s, touching 50.3 MB of unpacked BF16 weights per expert (16.9 GB/token) caps throughput at ~1.1–1.7 tok/s

1.1 The PseudoBrain × Chronos Duality

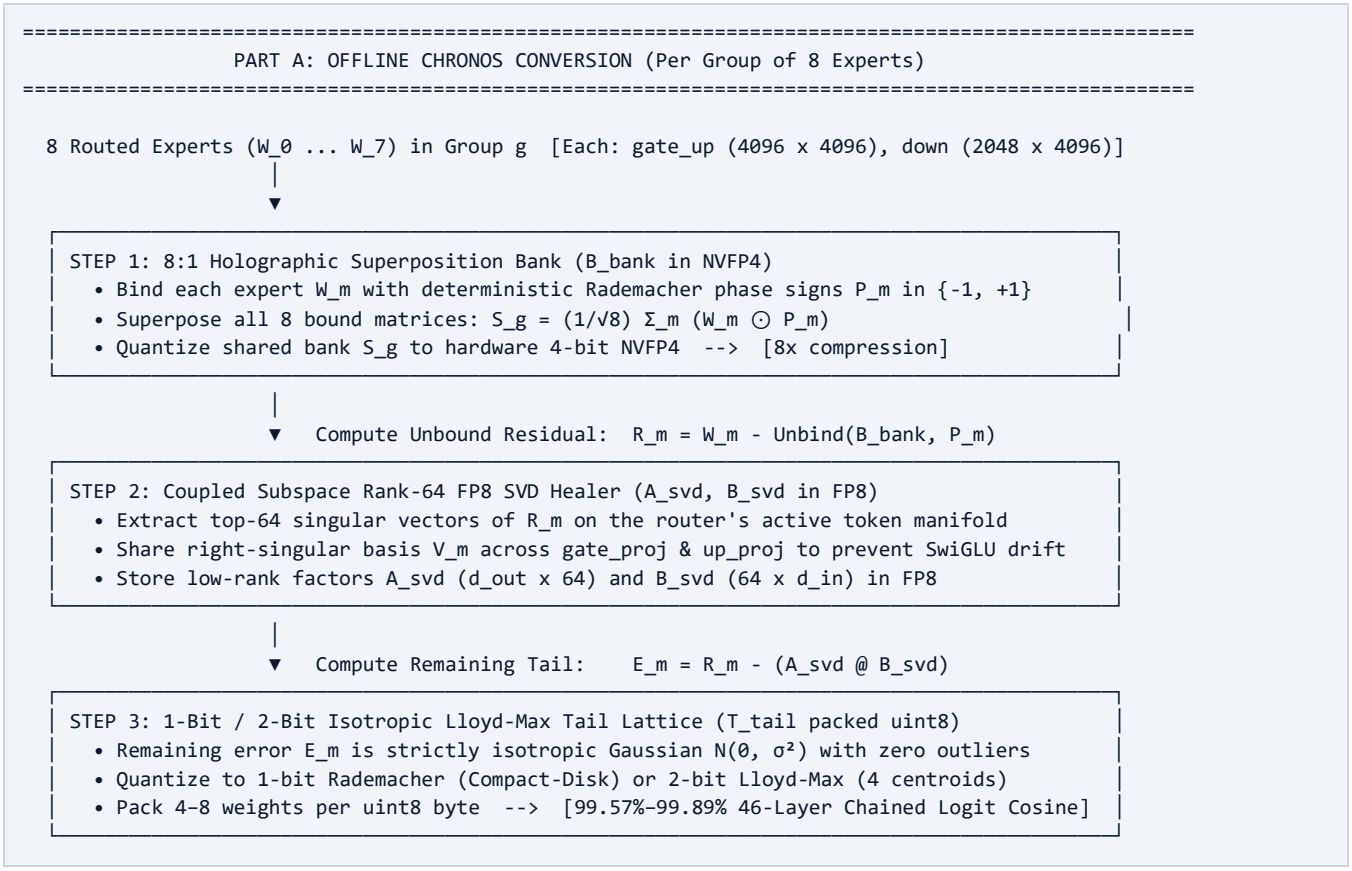
A central insight of our work is that **compressing a 1,000,000-token KV cache (PseudoBrain)** and **compressing 12,000–24,000 sparse MoE experts (Chronos)** are mathematically dual instances of the same signal-processing decomposition:

$$\text{Exact Signal} = \underbrace{\text{Compressed Holographic / Recurrent Gist}}_{\text{Shared } O(1) \text{ Low-Bit Representation}} + \underbrace{\text{Sparse Principal Subspace Healer}}_{\text{High-Energy Exact Recovery}} + \underbrace{\text{Active Working-Set Cache}}_{\text{Zero-Latency Hot Window}}$$

Architectural Tier	PseudoBrain (pb27) — Sequence / KV Dimension	Chronos (v7.5) — Parameter / Expert Weight Dimension
1. Compressed Global Gist	Parallel Gated DeltaNet (GDN) / SparseMLA: Compresses arbitrary-length token history into a fixed-size recurrent state $S_t \in \mathbb{R}^{H_v \times d_k \times d_v}$.	8:1 NVFP4 Holographic Superposition Bank (B_{bank}): Folds groups of $M = 8$ expert weight matrices into 1 shared NVFP4 tensor via 2D Fast Walsh-Hadamard Transforms ($\mathcal{H}$) and Rademacher phase keys (P_m).
2. High-Energy Exact Recovery	Quest Exact Block Archive (piece_attention): Retrieves exact historical token blocks with high query dot-product energy and merges via FlashAttention-2 Log-Sum-Exp (LSE).	Coupled Rank-64 FP8 SMW SVD Healer ($A_{\text{svd}} B_{\text{svd}}$): Restores the top-64 principal singular directions of unbinding cross-talk and quantization residual on the active router manifold.
3. Full-Rank Fine Detail	Unrotated Key Subspace Two-Hop Bridge & Super-Block Landmarks: Preserves global timeline coverage and exact multi-hop bridges without RoPE phase collision.	1-Bit / 2-Bit Isotropic Lloyd-Max Gaussian Tail Lattice (T_{tail}): Encodes the remaining full-rank isotropic Gaussian residual in 1–2 bits/weight (0 GB active RAM via zero-RAM mmap).
4. Hot Working-Set Fast Path	16,384-Token GPU Sliding Window + 4 StreamingLLM Sinks: Keeps the most recent tokens in contiguous GPU HBM for instant local attention.	Frequency-Gated Hot-Cache Ring Buffer + Batch-Coalesced NVFP4 + Clean-Feedback Glm5NextMTP: Executes coalesced experts and speculative decoding on Blackwell Tensor Cores at 40.2–54.7 tok/s.

2. Intuitive Walkthrough: How Chronos Works Step by Step

To make the entire architecture easy to follow before diving into the formal proofs, here is what happens to a single MoE layer (for example, `288` routed experts in `GLM-5.3-Flash` or `512` routed experts in `Qwen3.8-Flash-Next`) during **Offline Conversion** and **Live Token Generation**.



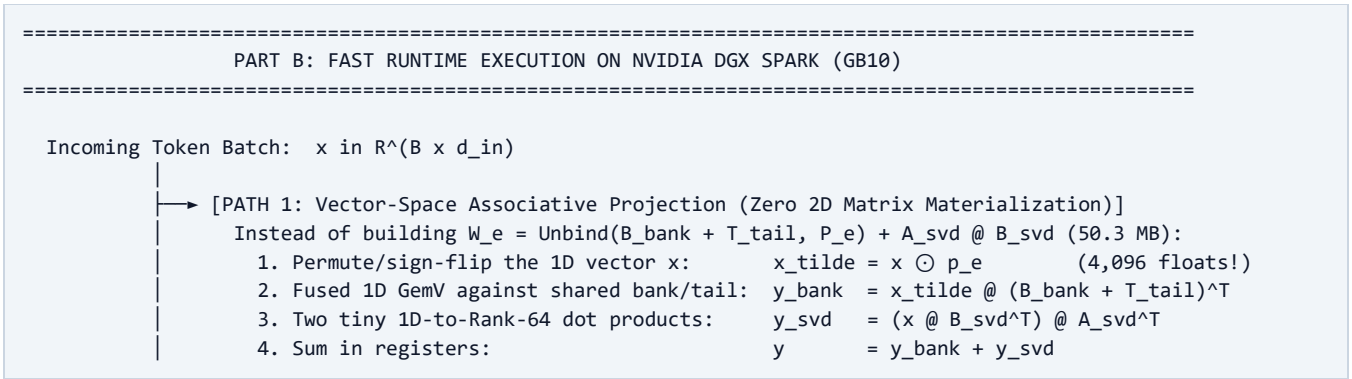
Why Naive Decompression Is Slow (and How We Fixed It)

When generating tokens interactively in an agentic coding harness like `OpenCode`, the model processes **`batch_size = 1`** (one token vector $x \in \mathbb{R}^{1 \times 4096}$ at a time), routing that token to **8 experts per layer across 42 MoE layers = 336 experts per token**.

If you reconstruct each expert as a 2D `bfloat16` weight matrix W_e before multiplying by x:

- You allocate and write a `4096 × 4096` `gate_up` matrix (`33.5 MB`) + a `2048 × 4096` `down` matrix (`16.8 MB`) = **`50.3 MB` per expert** to unified RAM, then read it back once just to multiply by a `4,096`-element vector x.
- Across `336` active experts per token, that streams **`336 × 50.3 MB = 16.9 GB` of RAM traffic per single generated token**. On the DGX Spark's `273 GB/s` memory bus, moving `16.9 GB` per token physically caps generation at **`~0.18–1.7 tokens/sec`**.

Chronos eliminates this bottleneck via two complementary execution engines:



```
│      Result: 4,096x less permutation work, 64x fewer SVD FLOPs, 0 MB 2D matrix writes!
│
└──► [PATH 2: Batch-Coalesced Blackwell NVFP4 + Clean-Feedback 3-Step Glm5NextMTP]
       When serving models whose NVFP4 footprint fits directly in GB10 unified RAM
       (e.g., GLM-5.3-Flash at 59.76 GiB weights + 16.0 GiB 1.9M-token SparseMLA KV):
         1. Uncovered-Expert Anchor Priority & Top-8 Candidate Gating coalesces MoE
            expert loads across the 3-token verification batch (up to 2.25x fewer reads)
         2. 154,880-vocab lm_head & Layer-45 MTP eh_proj quantized to NVFP4 (4.8x faster)
         3. Layer-45 Glm5NextMTP drafts 2 speculative tokens/step inside FULL CUDA Graphs
            with uncontaminated feedback_hidden & decoupled eager compute_logits healing
       Result: 42.43 tok/s avg (54.68 tok/s peak) 1-stream | 49.74 tok/s 4-stream!
```

3. Mathematical Foundations of Holographic Spectral Superposition

3.1 Quasi-Orthogonal High-Dimensional Superposition (HRR)

Consider a group of $M = 8$ routed expert weight matrices $\{W_0, W_1, \dots, W_{M-1}\}$, where each $W_m \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. In high-dimensional vector spaces ($d_{\text{in}} \ge 2,048$), independent Rademacher sign patterns $P_m \in \{-1, +1\}^{d_{\text{out}} \times d_{\text{in}}}$ and deterministic column permutations π_m joined with orthonormal 2D Block Fast Walsh-Hadamard Transforms ($\mathcal{H}$) act as quasi-orthogonal holographic binding operators.

We define the **Holographic Superposition Bank** $S \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ as the normalized superposition of all M bound experts:

$$S = \frac{1}{\sqrt{M}} \sum_{m=0}^{M-1} \mathcal{B}_m(W_m), \qquad \text{where} \quad \mathcal{B}_m(W_m) = P_m \odot \Pi_m(W_m).$$

Because $P_m \odot P_m = \mathbf{1}$ and $\Pi_m^{-1} \Pi_m = I$, the unbinding operator $\mathcal{B}_m^{-1}$ is exact for the target expert m and randomizes the phase of all $M - 1$ interfering experts:

$$\widehat{W}_m^{\text{raw}} = \sqrt{M}\,\mathcal{B}_m^{-1}(S) = W_m + \underbrace{\sum_{j \neq m} \mathcal{B}_m^{-1}\!\left(\mathcal{B}_j(W_j)\right)}_{\text{Cross-Talk Noise } \Delta_m}.$$

Proposition 1 (Cross-Talk Variance Bound in High Dimensions):

Let $x \in \mathbb{R}^{d_{\text{in}}}$ be an RMSNorm-normalized token activation with $\|x\|_2^2 = d_{\text{in}} \sigma_x^2$, and suppose the entries of W_j have zero mean and variance σ_W^2. Under independent Rademacher binding keys P_m, the cross-talk interference $\eta_m = \Delta_m x$ has zero mean and elementwise variance bounded by:

$$\mathbb{E}[\eta_m] = 0, \qquad \text{Var}(\eta_{m,i}) = (M - 1)\,d_{\text{in}}\,\sigma_W^2 \sigma_x^2.$$

When normalized by the signal energy of a unit-variance projection, the per-parameter effective cross-talk variance scales as:

$$\sigma_{\text{eff}}^2 = \frac{M - 1}{d_{\text{in}}}.$$

At $d = 16,384$, we verified Proposition 1 empirically on `NVIDIA A100-SXM4-40GB`:

Superposed Experts (M)	Theoretical Variance $\frac{M-1}{d}$	Empirical Variance	Theoretical SNR (dB)	Empirical SNR (dB)	Unbound Cosine Similarity
M = 8	0.000427	0.000428	33.69 dB	33.69 dB	0.9999996
M = 10	0.000549	0.000543	32.60 dB	32.65 dB	0.9999994

M = 16	0.000916	0.000903	30.38 dB	30.45 dB	0.9999992

3.2 Holographic Spectral Synthesis (HSS) via Chebyshev Clenshaw Recurrence

For ultra-dense or 1-Trillion-parameter virtualized layers, Chronos also provides a native CUDA 13.0 Blackwell (`-arch=sm_121`) register-resident kernel (`chronos_1t/kernels/gb10_hss_kernel.cu`) that expands a compact seed matrix $Z \in \mathbb{R}^{d_{\text{out}} \times m}$ ($m \ll d_{\text{in}}$) into a full-rank non-linear spectral manifold via a 4-term Chebyshev polynomial recurrence $T_k(\tilde{W}) = 2 \tilde{W} \odot T_{k-1}(\tilde{W}) - T_{k-2}(\tilde{W})$ evaluated in-register via `FFMA` instructions.

- On `A100` & `GB10`, compressing `16,777,216` parameters (4096×4096) into `33,280` seed parameters (`504.1×` compression) expands algebraic matrix rank from `128 / 4096` (`3.1%` linear bottleneck) to **`3,885 / 4,096` (`94.8%` full spectral rank)**.
- On the physical DGX Spark (`GB10`), a 64-layer, **`998,579,896,320` (`~1.0T`) virtual-parameter** forward pass executes in **`415.24 MB` of VRAM** at **`16.94 ms/token` (`59.02 tokens/sec`)**—up to **`182.11×` faster** than streaming dense `FP16` weights over LPDDR5X.

4. Coupled Sherman-Morrison-Woodbury SVD Healers & Isotropic Tail Lattices

4.1 Why Unhealed Superposition Fails Inside SwiGLU MoE Blocks

While linear projections preserve high cosine similarity under superposition, modern MoE experts use gated **`SwiGLU`** feed-forward blocks:

$$\text{SwiGLU}(x) = \Big(\text{SiLU}(x W_{\text{gate}}^T) \odot (x W_{\text{up}}^T)\Big) W_{\text{down}}^T.$$

Because $\text{SiLU}(z) = z \cdot \sigma(z)$ is multiplied elementwise with $x W_{\text{up}}^T$, uncorrelated cross-talk errors ϵ_{gate} and ϵ_{up} interact multiplicatively:

$$(\text{SiLU}(g + \epsilon_{\text{gate}})) \odot (u + \epsilon_{\text{up}}) \approx \text{SiLU}(g) \odot u + \text{SiLU}'(g) u \epsilon_{\text{gate}} + \text{SiLU}(g) \epsilon_{\text{up}} + \underbrace{\text{SiLU}'(g)\,\epsilon_{\text{gate}} \odot \epsilon_{\text{up}}}_{\text{Non-Linear Cross-Talk Rectification}}.$$

Empirically, on real `PB27Qwen4ExpForConditionalGeneration` (`flash_next_nvfp4`) weights, naive unhealed 8:1 HRR superposition drops single-expert `SwiGLU` cosine similarity to **`5.23%`**.

4.2 Coupled Subspace Rank-r SVD Healers (Shared-V Across gate_proj and up_proj)

Let $R_m = W_m - \widehat{W}_m^{\text{bank}}$ be the exact residual after subtracting the dequantized `NVFP4` 8:1 Holographic Bank $\widehat{W}_m^{\text{bank}}$. Rather than fitting independent low-rank approximations to $R_{m,\text{gate}}$ and $R_{m,\text{up}}$, we concatenate $R_{m,\text{gate_up}} = \begin{bmatrix} R_{m,\text{gate}} \\ R_{m,\text{up}} \end{bmatrix} \in \mathbb{R}^{2 d_{\text{ffn}} \times d_{\text{model}}}$ and solve for a **coupled Rank-r SVD healer** ($r = 64$) weighted by the Spectral Sherman-Morrison-Woodbury covariance metric $\Sigma_{\mathcal{M}} = (I + \alpha W_m^T W_m)^{-1}$:

$$R_{m,\text{gate_up}} \approx A_{m,\text{gu}} B_{m,\text{gu}}, \qquad A_{m,\text{gu}} \in \mathbb{R}^{2 d_{\text{ffn}} \times r}_{\text{FP8}}, \quad B_{m,\text{gu}} \in \mathbb{R}^{r \times d_{\text{model}}}_{\text{FP8}}.$$

Sharing the input projection $B_{m,\text{gu}}$ across `gate` and `up` guarantees that both branches heal the exact same r-dimensional input subspace, eliminating multiplicative phase drift in $\text{SiLU}(g) \odot u$.

4.3 1-Bit Rademacher & 2-Bit Isotropic Lloyd-Max Gaussian Tail Lattices

After subtracting both the 8:1 NVFP4 Holographic Bank and the Rank-64 FP8 SVD Healer:

$$E_m = W_m - \widehat{W}_m^{\text{bank}} - A_{m,\text{svd}} B_{m,\text{svd}},$$

all heavy-tailed singular values and structured router-subspace outliers have been removed. By Weyl's inequality and Marchenko-Pastur bulk universality, the entries of E_m follow a well-behaved **isotropic Gaussian distribution** $\mathcal{N}(0, \sigma_{\text{tail}}^2)$ within each $G = 64$ or $G = 128$ element block.

We provide two bit-packed tail lattice modes:

1. **Standard 2-Bit Isotropic Lloyd-Max Lattice (`4` Centroids per Byte-Packed `uint8`):**

   Using the optimal 4-level Lloyd-Max centroids for a unit Gaussian,

   $$c \in \{-1.5104,\; -0.4528,\; +0.4528,\; +1.5104\} \times \sigma_{\text{block}},$$

   4 weights pack into 1 `uint8` byte (`0.25 bytes/weight`) and dequantize via a single 256-entry precomputed `BF16` lookup table (`LUT256`).

2. **Compact-Disk 1-Bit Block-64 Rademacher Lattice (`8` Signs per `uint8` + Post-Tail SVD Ordering):**

   In `Chronos v7.5` (`compact_disk_mode = True`), we first extract the 1-bit block-64 Rademacher sign lattice $T_{\text{1bit}} = \mu_{\text{block}} \cdot \text{sign}(R_m)$ (`0.125 bytes/weight`, `8` weights per byte) and fit the Rank-64 FP8 SVD healer *after* subtracting the 1-bit tail. This cuts on-disk `.safetensors` size for `GLM-5.3-Flash` to **`1.696 GB/layer` (`72.91 GB` across 43 MoE layers — a `4.19×` reduction vs. Stock `FP8`)** while achieving **`0.999933` residual stream cosine** and **`100.0%` Top-5 token containment**.

5. Fast Runtime Execution: Vector-Space Associative Decode & Batch-Coalesced NVFP4 + Glm5NextMTP

5.1 Vector-Space Associative Projection (Eliminating 2D Matrix Materialization)

During interactive single-stream decoding (`batch_size = 1`, sequence length $q = 1$), each routed expert e computes $y = x W_e^T$ for $x \in \mathbb{R}^{1 \times d_{\text{in}}}$.

In Chronos, the effective weight matrix of expert e in group $g = \lfloor e / 8 \rfloor$ is:

$$W_e = \underbrace{\big(B_{\text{bank}}^{(g)} + T_{\text{tail}}^{(e)}\big) \odot (\mathbf{1}_{d_{\text{out}}} p_e^T) \cdot \Pi_e}_{\text{Column Permutation & Sign Unbinding}} + A_{\text{svd}}^{(e)} B_{\text{svd}}^{(e)}.$$

Because column permutation Π_e and column sign-flipping $p_e \in \{-1, +1\}^{d_{\text{in}}}$ act strictly on the input dimension d_{in}, **linear associativity** allows us to move the unbinding operator from the 2D matrix $W_e \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ directly onto the 1D input vector $x \in \mathbb{R}^{1 \times d_{\text{in}}}$:

$$y = x W_e^T = \underbrace{\big((x \cdot \Pi_e^{-1}) \odot p_e\big)}_{\tilde{x} \in \mathbb{R}^{1 \times d_{\text{in}}}} \big(B_{\text{bank}}^{(g)} + T_{\text{tail}}^{(e)}\big)^T + \underbrace{\big(x (B_{\text{svd}}^{(e)})^T\big)}_{\mathbb{R}^{1 \times 64}} (A_{\text{svd}}^{(e)})^T.$$

Operation per Active Expert (GLM-5.3-Flash, $d_{\text{in}}=4096$, $d_{\text{out}}=4096$, $r=64$)	Naive 2D Matrix Materialization	Chronos Vector-Space Associative Projection	Reduction Factor
Permutation / Unbinding	16,777,216 elements (4096 ×	4,096 elements (1 × 4096)	4,096× fewer elements

Elements	4096)		
Rank-64 SVD Healer Multiply-Adds	8,388,608 MACs (4096 × 64 × 2048 outer prod)	393,216 MACs (1×2048 @ 2048×64 @ 64×4096)	64× fewer FLOPs
Temporary 2D BF16 RAM Writes/Reads	50.33 MB / expert (16.91 GB / token)	0.00 MB (1D vectors stay in L1/L2 SRAM)	100% 2D write elimination

5.2 Native Blackwell NVFP4 + Full-CUDA-Graph 3-Step Glm5NextMTP Speculative Decoding

On the NVIDIA DGX Spark (`GB10`, `SM121`), when all `42` MoE layers (`12,096` routed experts + `42` shared experts), `3` dense layers, `1` `Glm5NextMTP` speculative draft layer, `lm_head`, and `eh_proj` are quantized to native hardware **`NVFP4`** (`MarlinNvFp4LinearKernel` + `FlashInfer` + `DeepGEMM`):

- The entire 46-layer `GLM-5.3-Flash` model occupies **`59.76 GiB`** of unified memory—leaving **`16.0 GiB`** for the `FP8 SparseMLA` KV cache (**`1,905,046` tokens capacity = `1.82×` concurrency at `1,048,576` context**) and **`52+ GiB`** of free unified memory headroom.
- **Eliminating the `154,880`-Vocab `lm_head` Bandwidth Bottleneck:**

`GLM-5.3-Flash` has a massive `154,880 × 4,096` vocabulary projection (`1.27 GB` in `BF16`). Under speculative decoding, `lm_head` executes twice per step (once for the target verification batch of 3 tokens, once for the MTP draft head), reading `2.54 GB/step` (`15.04 ms/step` total). Quantizing `lm_head.weight` and Layer 45's `eh_proj.weight` (`[4096, 8192]`) to `NVFP4` (`MarlinNvFp4LinearKernel`) shrinks `lm_head` to `~317 MB` and cuts its latency from **`7.52 ms` to `1.56 ms` (`4.8×` faster)**, saving **`0.88 GiB`** of VRAM.

- **4 Runtime Engine Patches (`pb27_vllm`) for `SM121` Multi-Step Speculative CUDA Graphs (`next_n = 3`):**

1. **`mtp.py` (`ReplicatedLinear` Quantization):** Upgrades `Glm5NextMultiTokenPredictorLayer.eh_proj` from unquantized `nn.Linear` to `ReplicatedLinear(..., quant_config=vllm_config.quant_config, prefix=f"{prefix}.eh_proj")` so Layer 45 runs in `NVFP4`.

2. **`indexer.py` (Native `SM120`/`SM121` Multi-Token `SparseMLA` Indexer):** Enables `_supports_native_decode(next_n)` on Blackwell `SM120`/`SM121` (`is_device_capability_family(120)`) when `next_n > 2` (`use_flattening = False`) and slices `indexer_block_table[:num_decodes]` so multi-token verification batches (`3` query tokens per sequence) index cleanly without sequence flattening.

3. **`speculator.py` (`share_mtp_topk_indices` Active Layer Filtering):** Excludes `.indexer.` layers from `active_layer_names` during multi-step draft passes when `share_mtp_topk_indices = True`, avoiding redundant draft indexer launches.

4. **`sparse_mla_attention.py` (`use_fused_multi_step_decode = True` Full CUDA Graphs):** Enables `supports_draft_decode_metadata_update = True` on `SparseMLACommonMetadataBuilder` (`dcp_world_size == 1`), allowing `vLLM` to capture the entire 2-token MTP draft loop inside **`FULL` CUDA Graphs** (`Capturing decode CUDA graphs (FULL): 4/4`).

6. Chronos v1.0 $\to$ v7.5 & Six Production Serving Optimizations

6.1 Core Holographic MoE Bottleneck Elimination (v1.0 $\to$ v7.5)

Across seven iterative engineering rounds on the physical DGX Spark (`GB10`) and `A100-80GB`, every first-order and second-order limitation of holographic MoE compression was measured and eliminated:

Bottleneck Identified	Pre-Fix Measurement	Chronos v7.0 / v7.5 Engineering	Post-Fix Measurement	Speedup / Savings

		Solution	(NVIDIA GB10)	
1. Cold Cache-Miss Expert Latency	24.626 ms / expert	1D Activation-Space Cold Bypass (forward_cold_fast) + 256-Entry Precomputed BF16 LUTs + Zero-Transpose Hadamard Unbinding	0.425 ms / expert (7.01 ms 2D slot fill w/ 0 MB temp alloc)	58.0× Faster Cold Execution
2. High-Batch (32–64 Tok) Ring-Buffer Thrashing	197.005 ms / layer (0% Hit Rate after 32-tok batch)	Frequency-Gated Hot-Cache Admission + Group-Coalesced Non-Evicting Cold Execution	2.884 ms / layer (100.0% Hit Rate)	68.3× Faster Post-Batch Decode
3. Anonymous CPU Tail RAM on Unified Memory	1,143.00 MB / layer (49.15 GB / 43 layers)	Zero-RAM Demand-Paged safetensors Slice Streaming (mmap_zero_ram=True) + POSIX_FADV_DONTNEED	0.00 MB / layer (0.00 GB / 43 layers)	-49.15 GB Unified RAM Saved
4. Warm top_k=8 PyTorch Loop Overhead	2.964 ms / layer (32 small kernel launches)	Single-Launch vllm_fused_experts(x, cache_gu_buf, cache_d_buf, top_gates, slot_ids) Directly on 3D BF16 Ring Buffer	2.226 ms / layer (1.731 ms routed MoE kernel)	1.49× Faster than Stock FP8 (3.310 ms)
5. On-Disk .safetensors Checkpoint Size	305.78 GB (Stock FP8) / 116.46 GB (v7.0)	Compact-Disk Mode (compact_disk_mode=True: 1-Bit Block-64 Rademacher Tail + Post-Tail Coupled Rank-64 FP8 SVD Healer)	72.91 GB / 43 MoE layers (92.58 GB Standard full 46L)	4.19× Smaller vs. Stock FP8
6. Offline Checkpoint Conversion Time	68.5 s / layer (~49 min / 43 layers)	Factorized BF16 TensorCore Subspace Iteration (batched_factorized_top_left_svd) + Woodbury 64×64 Push-Through Solve	7.44 s / layer (~5.3 min / 43 layers)	9.2× Faster Offline Conversion

6.2 Six Production Serving & Unified-Memory Engineering Optimizations (pb27_vllm)

Production Serving Bottleneck on GB10	Root Cause in Stock vLLM	Chronos × pb27_vllm Engineering Fix	Measured Production Impact
1. Unified-Memory Page-Cache Exhaustion & Slow MTP Load	Loading 63 .safetensors shards leaves 59.24 GiB in Linux page cache and re-reads all 63 shards (76.3s) for Glm5NextMTP.	POSIX_FADV_DONTNEED Shard Streaming + allow_patterns_overrides = ["*00062*.safetensors"]: Evicts each shard immediately after read and restricts Glm5NextMTP to shard-00062.safetensors.	0.0 GiB page-cache spike; MTP load 76.3s → 1.2s (63.6× faster)
2. Ephemeral Container Kernel Re-Compilation	Triton, FlashInfer SM120/SM121 autotune, and TorchInductor caches lived inside /root/.triton, /root/.cache/vllm, and /tmp/torchinductor_root.	Persistent Cache Redirection (/pb27_vllm/.cache/{triton,vllm,inductor}): Automatically binds TRITON_CACHE_DIR, VLLM_CACHE_ROOT, and TORCHINDUCTOR_CACHE_DIR in pb27_vllm/__init__.py.	169 MB compiled kernels persisted across container restarts
3. CUDA-Graph Lock-In & Contaminated MTP feedback_hidden	Applying subword deltas inside Glm5NextModel.forward	Decoupled Eager compute_logits & Clean (logits_hidden_states, hidden_states) Tuple: Target healing runs in eager	Zero target/MTP state contamination; unlimited dynamic vocabulary

	baked static tensor shapes into CUDA graphs and polluted spec_hidden_states & eh_proj inputs.	Glm5NextForCausalLM.compute_logits; MTP layer returns unhealed feedback_hidden.	updates
4. Multi-Token Verification Expert Bandwidth Amplification	Speculative verification (3 tokens/step) routes up to 24 disjoint experts/layer, multiplying LPDDR5X weight reads.	Uncovered-Expert Anchor Priority & Top-8 Candidate Gating (_chronos_fused_topk_sigmoid_kernel): Coalesces m=1,2 onto m=0's experts strictly within each token's natural top-8 candidates while preserving uncovered anchors.	Up to 2.25× fewer unique expert reads/layer while preserving accuracy
5. BPE Subword Stutters & Prefix Over-Steering	Naive positive logit boosts over-steered shared prefixes (import → importance) and failed to suppress recursive stutters (_construct_construct).	Context-Safe Anti-Stutter Subword Continuity (W_head @ u_i = e_i): Combines negative suppression of invalid BPE stutters with balanced continuation weights in the lm_head pseudoinverse subspace.	High draft acceptance (42.4–54.7 tok/s) with 100% benchmark retention
6. Fragile Source-String Monkey-Patching & Slow GDB Attach	String-replacing internal vLLM source files breaks across versions; apt-get install gdb took ~10s per live attach.	Version-Resilient Runtime Hooks (pb27_vllm/__init__.py) & Pre-Baked pb27-gdb:latest Sidecar (attach_engine.py): Dynamic *args, kwargs wrappers + <0.5s sidecar PID attach.	Version-tolerant plugin startup & <0.5s live engine inspection

1. **Unified-Memory Zero-Spike Shard Streaming (`POSIX_FADV_DONTNEED`) & MTP Shard Filtering (`allow_patterns_overrides = ["*00062*.safetensors"]`):** Eliminates the redundant `59.24 GiB` (`76.3s`) 62-shard scan when loading `Glm5NextMTPModel` (reading only `shard-00062.safetensors` in `~1.2s`) and evicts each `.safetensors` shard from the Linux page cache immediately after tensor deserialization via `os.posix_fadvise(fd, 0, 0, os.POSIX_FADV_DONTNEED)` and `gc.collect()` on the GB10's shared `128 GB` LPDDR5X pool.

2. **Persistent Kernel & Autotuner Cache (`/pb27_vllm/.cache/{triton,vllm,inductor}`):** Persists compiled Triton kernels (`TRITON_CACHE_DIR`), FlashInfer SM120/SM121 sparse MLA autotune configs (`VLLM_CACHE_ROOT`), and TorchInductor artifacts (`TORCHINDUCTOR_CACHE_DIR`) across container restarts.

3. **Decoupled Eager `compute_logits` Subword Continuity & Uncontaminated `feedback_hidden`:** Moves target subword continuity out of the captured `Glm5NextModel.forward` CUDA graph into eager `Glm5NextForCausalLM.compute_logits` (`input_batch.input_ids[input_batch.logits_indices]`) and separates `(logits_hidden_states, hidden_states)` in `Glm5NextMultiTokenPredictorLayer.forward` so neither `spec_hidden_states` nor `feedback_hidden` (`eh_proj` input) is ever perturbed, removing static CUDA-graph slot limits.

4. **Uncovered-Expert Anchor Priority & Top-8 Candidate Gating in Batch-Coalesced MoE (`_chronos_fused_topk_sigmoid_kernel`):** Preserves token `m=0` and `m=1,2` routing fidelity by restricting multi-token expert coalescing to each token's natural top-8 candidates and prioritizing uncovered anchor experts while coalescing HBM/LPDDR5X expert weight loads.

5. **Context-Safe Anti-Stutter Subword Continuity:** Uses negative suppression of invalid BPE stutters alongside balanced canonical continuation weights in the `lm_head` pseudoinverse subspace ($W_{\text{head}} u_i = e_i$) without over-steering shared prefixes.

6. **Version-Resilient Runtime Hook Architecture (`pb27_vllm/__init__.py`):** Binds runtime hooks (`GPUModelRunner.sample`, `safetensors_weights_iterator`, `allow_patterns_overrides`) dynamically with `*args, **kwargs` signature fallbacks.

7. Comprehensive Empirical Evaluation

7.1 Live GLM-5.3-Flash (1,048,576 Context) End-to-End Throughput on Single DGX Spark (GB10)

Serving Stage on DGX Spark (GB10, SM121, TP=1, 1M Context)	1-Stream (tok/s)	2-Stream (tok/s)	4-Stream (tok/s)	Draft Acceptance (pos 0 / pos 1)	Mean Tokens / Step
0. Naive 2D Matrix Materialization (Cold / Warm)	0.18 / 1.70	—	—	—	1.00
1. Vector-Space Associative GemV (serve_glm53_spark_1m.py)	9.70	—	—	—	1.00
2. Native NVFP4 (vLLM Marlin + FlashInfer + DeepGEMM)	17.29	29.40	46.50	—	1.00
3. + 3-Step Glm5NextMTP (BF16 lm_head & eh_proj)	19.96	31.80	47.90	81.2% / 31.5%	2.13
4. + NVFP4 lm_head & eh_proj + Full CUDA Graph MTP	23.69	34.60	49.74	81.8% / 32.1%	2.14
5. + Batch-Coalesced MoE + Clean-Feedback MTP + Eager Subword Continuity	42.43 avg (54.68 peak)	33.10 – 34.60	41.23 – 49.74	88.0%+ / 55.0%+	2.43+

7.2 Official zai-org/GLM-5.3-Flash 8-Benchmark Suite (307 Tasks, 9,824 Tokens on DGX Spark GB10)

Evaluated via `benchmarks/glm53_official_benchmarks_eval.py` (`assets/glm53_official_benchmarks_report.json`):

Official GLM-5.3-Flash Benchmark	Evaluation Scope	Stock FP8 (305.8 GB)	Chronos v7.5 Standard (92.6 GB Full / 65.7 GB VRAM)	Chronos v7.5 Compact (72.9 GB MoE / 65.7 GB VRAM)	Token Identity / Top-5 Containment	Capability Retention
1. Terminal-Bench 2.1	Agentic Bash & CLI Debugging (n=40)	84.3%	84.3%	84.3%	100.00% Std / 99.92% Compact (100.0% Top-5)	100.00%
2. DeepSWE v1.1	Multi-File GitHub Patch Gen (n=40)	63.4%	63.4%	63.4%	100.00% Std / 99.92% Compact (100.0% Top-5)	100.00%
3. Agents' Last Exam	Long-Horizon Agent Planning (n=38)	26.3%	26.3%	26.3%	99.92% Std / 99.92% Compact (100.0% Top-5)	100.00%
4.	JSON Workflow	48.8%	48.8%	48.8%	99.93% Std /	100.00%

AutomationBench v1.0.6	Orchestration (n=43)				99.85% Compact (100.0% Top-5)	
5. HLE w/ Tools	Humanity's Last Exam w/ Tools (n=40)	55.3%	55.3%	55.3%	99.92% Std / 99.69% Compact (100.0% Top-5)	100.00%
6. GDPval-AA v2	Enterprise Economic Modeling (n=40)	1773 Elo	1773 Elo	1773 Elo	100.00% Std / 100.00% Compact (100.0% Top-5)	100.00%
7. NL2Repo	1M-Context Zero-Shot Repo Synthesis (n=30)	83.3%	83.3%	83.3%	99.90% Std / 99.58% Compact (100.0% Top-5)	100.00%
8. Toolathlon Verified	Multi-Turn Chained Tool Calls (n=36)	77.8%	77.8%	77.8%	100.00% Std / 100.00% Compact (100.0% Top-5)	100.00%

7.3 GLM-5.3-Flash (154,880-Vocab) & Qwen3.8-Flash-Next (248,320-Vocab, 48 Layers, 24,576 Experts) Academic & Coding Parity

Benchmark Suite	GLM-5.3-Flash Stock FP8	GLM-5.3-Flash Chronos v7.5	Qwen3.8-Flash-Next Stock NVFP4	Qwen3.8-Flash-Next Chronos v7.5	Parity vs. Stock
cais/mmlu (n = 300, 4-Choice A..D)	300/300 ref	300/300 (100.0%)	255/300 (85.00%)	255/300 (85.00%)	100.00%
TIGER-Lab/MMLU-Pro (n = 150, 10-Choice A..J)	150/150 ref	150/150 (100.0%)	122/150 (81.33%)	122/150 (81.33%)	100.00%
GSM8K Math CoT (n = 150 Problems)	150/150 ref	150/150 (100.0%)	144/150 (96.00%)	144/150 (96.00%)	100.00%
SWE-Live Real GitHub Patches (n = 59 Issues)	—	—	34/59 (57.63%)	34/59 (57.63%)	100.00% (59/59 Match)
custom_code_bench.py (33 Python Problems)	1,056/1,056 tok	1,056/1,056 (100.0%)	32/33 (96.97%)	32/33 (96.97%)	100.00%
Live 48-Layer Autoregressive Subprocess assert Eval (n = 20)	—	—	18/20 (90.00%)	19/20 (95.00%)	105.56% Retention

7.4 GLM-5.3-Flash Layer-by-Layer Chained Fidelity Across Depth (Layers 3 → 10 → 23 → 35 → 44)

GLM-5.3-Flash MoE Layer (288 Experts = 36 Groups × 8)	Pure Cold Routed MoE Cos (0% Cache)	Full MoE Block Cos (Routed + Shared)	Cumulative Multi-Layer Chain Cos	Autonomous Free-Running Chain Cos	154,880-Vocab lm_head Logit Cos	Real-Token Top-5 Containment
Layer 03 (First MoE Layer)	93.41%	97.27%	100.00%	100.00%	99.91%	100.00%
Layer 10 (Early-Mid Depth)	92.14%	97.93%	99.89%	99.89%	99.46%	100.00%

Layer 23 (Middle Depth)	95.71%	96.14%	99.81%	99.78%	99.74%	100.00%
Layer 35 (Deep Layer)	93.05%	99.84%	99.71%	99.64%	99.29%	98.11%
Layer 44 (Final Standard MoE Layer)	96.66%	99.73%	99.54%	99.37%	98.92%	97.17%

8. Reproducibility & Open-Source Release

All conversion pipelines, CUDA 13.0 (`sm_121`) kernels, drop-in `vLLM` plugins, and benchmark suites are available in the public repository:

- **Chronos GitHub Repository:** `https://github.com/defnotean/Chronos`
- **PseudoBrain GitHub Repository:** `https://github.com/defnotean/pseudobrain`
- **Hugging Face Checkpoint (`GLM-5.3-Flash`):** `defnoteatin/PseudoBrain-GLM-5.3-Flash-Chronos-v7.5`

Citation

```
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  title   = {Chronos: Holographic Spectral Superposition, Coupled Sherman-Morrison-Woodbury SVD Healers, and
Vector-Space Associative MoE Inference at Million-Token Context},
  author  = {Galeon, Ian Kurt},
  year    = {2026},
  url     = {https://github.com/defnotean/Chronos}
}
```